HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE423: Simulation and Modeling

Batch: 21st | Semester: Spring 2025

Time: 1 Hour 30 Minutes

Date: 25/06/2025 | **Mid-term Examination** | Total Marks: 30

Answer any **five** questions of the followings. $5 \times 6 = 30$

1. (a) Define any three terminologies of system simulation with example. 3
   (b) Briefly explain verification and validation in modeling. 3

2. (a) Explain Event-Driven modeling with relevant diagram. 3
   (b) List and define the primary performance measures of queueing systems. 3

3. (a) Briefly explain different types of perspectives for modeling of a system. 3
   (b) Describe Interaction Modeling with example. 3

4. (a) Define system and system environment in simulation study. 2
   (b) Draw the flow of control for the next event time advance approach. 4

5. (a) How does probability involved in discrete event simulation? 2
   (b) State the steps of discrete event simulation model. 4

6. (a) Differentiate between continuous and discrete systems. 2
   (b) Construct a diagram for arrival routine in queueing system. 4

7. (a) Define conceptual model with example. 2
   (b) Illustrate the four activities occur inside the single-server queueing system. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 469: Digital Image Processing

Batch: 21st — Semester: Spring 2025

Time: 1 hour 30 minutes

Date: 23/06/2025 — **Mid-Term Examination** — Total Marks: 30

Answer any **five** questions. **Marks: 5 x 6 = 30**

1. (a) Define image and Dynamic range of an image. 2
   (b) Find the Image and Range for the following function: 4
   $Y = x - 1$.

~~Sampling~~ Q~~uantization~~

2. (a) State the objectives and purposes of image processing. 3
   (b) Explain the classification of image processing on the basis of processing tasks. 3

3. (a) Mention some broad spectrum of image processing application. 2
   (b) Briefly explain the components of an image processing system. 4

4. (a) Draw the block diagram of a typical digital image processing sequence. 2
   (b) If an image of size 120x160 pixels is digitized by 1-bit, 4-bit and 8-bit quantization, Comment on the quality and storage requirements due to the quantization. 4

5. (a) Differentiate between brightness and luminance. 2
   (b) Mention and explain some features that are used in recognizing an object. 4

6. (a) To represent an image find the differences between 8-adjacency and m-adjacency with example. 3
   (b) Briefly explain the different types of adjacency to represent an image. 3

7. Explain the object determination process on human eye with figure. 6

noice, contrast, sharpening
segmentation, describing, classifying
understanding, making sense, recognizing
objects, performing tasks.

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 421: Software Engineering

Batch: 21st
Date: 21/06/2025

Mid-Term Examination

Semester: Spring 2025
Time: 1 Hour 30 Minutes
Total Marks : 30

Answer any **five** questions: 5 x 6 = 30

1. (a) Explain the general principles of software engineering. 3
   (b) Describe software application domains. 3

2. (a) Illustrate different types of process flow. 3
   (b) Describe three types of process pattern. 3

3. (a) Compare between Waterfall Model and Incremental Process Model. 3
   (b) Create a task set for a large and complex software project. 3

4. (a) Explain the human factors in agile process. 3
   (b) What is Agility? List all Agility principles. 3

5. (a) Draw a table for Extreme programming practices. 3
   (b) Describe Industrial Extreme Programming (IXP). 3

6. (a) Define product backlog and burn down chart in Scrum. 3
   (b) Define the role of a SCRUM master. 3

7. (a) Create a table comparing different agile techniques. 3
   (b) Briefly explain the stages involved in DevOps approach. 3

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Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE423: Simulation and Modeling

Batch: 21^{st} | Semester: Spring 2025

Time: 2 Hours

Date: 07-09-2025 | **Final Examination** | Total Marks: 40

Answer any **five** questions of the followings. Terminating - non-T $\qquad 5 \times 8 = 40$

1. (a) Explain the types of simulations with respect to output analysis with example. 4
   (b) Why is a histogram important in data distribution? How is a histogram constructed? 4

2. (a) Write the steps involved in the development of a useful model of input data. 4
   (b) Draw the diagram for event scheduling scheme in computer simulation. 4

3. (a) When is a stochastic process called a Continuous-Time Markov Chain? 3
   (b) Explain Markov Process with an example. 5

4. (a) List the issues that are considered in an accreditation decision 3
   (b) Explain inverse-transform technique of producing random variates for Exponential Distribution. ~~d = 0.2~~ 5

5. (a) What is Monte Carlo Simulation? What kind of input data is required for performing Monte Carlo simulations? 3
   (b) Use Monte Carlo Simulation to approximate the value of $\int_3^5 x^2 dx$ 5
   R1 → 0.21, 0.11, 0.71, 0.65, 0.41, 0.35, 0.17
   R2 → 0.96, 0.82, 0.56, 0.68, 0.58, 0.40, 0.82 $\qquad$ 28.42

6. Generate a sequence of 5 two-digit random numbers by employing the following methods:
   (i) Mixed Congruential Method 2
   (ii) Multiplicative Congruential Method 2
   (iii) Additive Congruential Method- $r_2 = (r_1 + r_0) \bmod 100$ 2
   The recursive equartion for the mixed congruential method is-
   $R_{i+1} = (21r_i + 53)$ (module 100) and seed $r_0=46$
   (iv) Midsquare Method (using same seed) $r_{i+1} = (r_i)^2$ 2

1. A sample of 100 arrivals of a customer at a retail sales depot is according to the following distribution: 8

Time between Arrival (min.)	Frequency
0.5	2
1.0	6
1.5	10
2.0	25
2.5	20
3.0	14
3.5	10
4.0	7
4.5	4
5.0	2

A study of the time required to service customers by adding up the bills, receiving payments and placing packages, yields the following distribution:

Time between Arrival (min.)	Frequency
0.5	12
1.0	21
1.5	36
2.0	19
2.5	7
3.0	5

Estimate the average percentage of customers waiting in time and average percentage of idle time of the server, by simulation, for the next 10 arrivals.

$R_1 = 27, 03, 88, 46, 61, 15, 94, 22, 74, 59$

$R_2 = 41, 77, 25, 90, 18, 35, 63, 08, 52, 80$

S start = Pre S end > Arr = Pre

S end = S start + S T

1 96%

41.17%

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 469: Digital Image Processing

Batch: 21st | Semester: Spring 2025

Final Examination | Time: 2 hours

Date: 13/09/2025 | Total Marks: 40

Answer any five questions. Marks: 5 x 8 = 40

1. (a) State the advantages and disadvantages of region growing of an image. 3
   (b) Briefly explain the Components in Pattern Recognition System of an image. 5

2. (a) State the edge thinning of an image with advantages. 2
   (b) Briefly explain the edge properties of an image. Independent, te 2
   (c) Briefly explain the working procedure of any one edge thinning algorithm. 4

3. (a) State the determining process of periodic noise by Frequency Domain Filtering of an image. 2
   (b) Briefly explain the different types of filtering of an image. 6

4. (a) Draw the degradation model with equation of an image. 2
   (b) Briefly explain different types of noise of an image. 6

5. (a) If R, G, B values of an image points are 140, 60 and 120 respectively, then find the H, S, L values of that image. 3
   (b) How to convert a color image to a grayscale image and then to a binary image? 5

6. (a) Briefly describe the two important linear point opearations for image enhancement. 3
   (b) Find the maximum, minimum, median and average filtering output of the bold mark pixel for the following image by using a 3x3 kernel: 5

0	250	200	100	10
0	220	10	120	0
50	80	150	90	45
90	70	210	240	0

$R = \frac{(G-B)}{max-min}$

$G = 2 + \frac{(B-R)}{max-min}$

$B = 4 + \frac{R-G}{max-mi}$

7. (a) Define Image moments in different form. Why moments are used? 3
   (b) What is an affine moment invariant? Prove that the moments $\mu_{01} = \mu_{10} = 0$. 5

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Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 421: Software Engineering

Batch: 21st | Semester: Spring 2025

Date: 10/09/2025 | Final Examination | Time: 2 Hours

Total Marks : 40

Answer any **five** questions: $5 \times 8 = 40$

1. (a) List the questions for validating requirements. 4
   (b) Explain how requirements model acts as a bridge between the system description and the design model. 4

2. (a) What is the purpose of a configuration model? Explain. 4
   (b) Briefly explain Navigation Modeling. 4

3. (a) Does refactoring mean that you modify the entire design iteratively? If not what does it mean? 4
   (b) Describe software quality guidelines and attributes. 4

4. (a) List the architectural genres for software based system. 4
   (b) Briefly explain the taxonomy of Architectural styles. 4

5. (a) Write about user interface design golden rules. 4
   (b) Explain four common design issues for a WebApp. 4

6. (a) Illustrate the WebApp design pyramid. 4
   (b) What is the most unusable website you have ever visited and why? 4

7. (a) Describe four activities that help a software team to achieve high software quality. 4
   (b) Explain software quality dilemma in your own words. 4